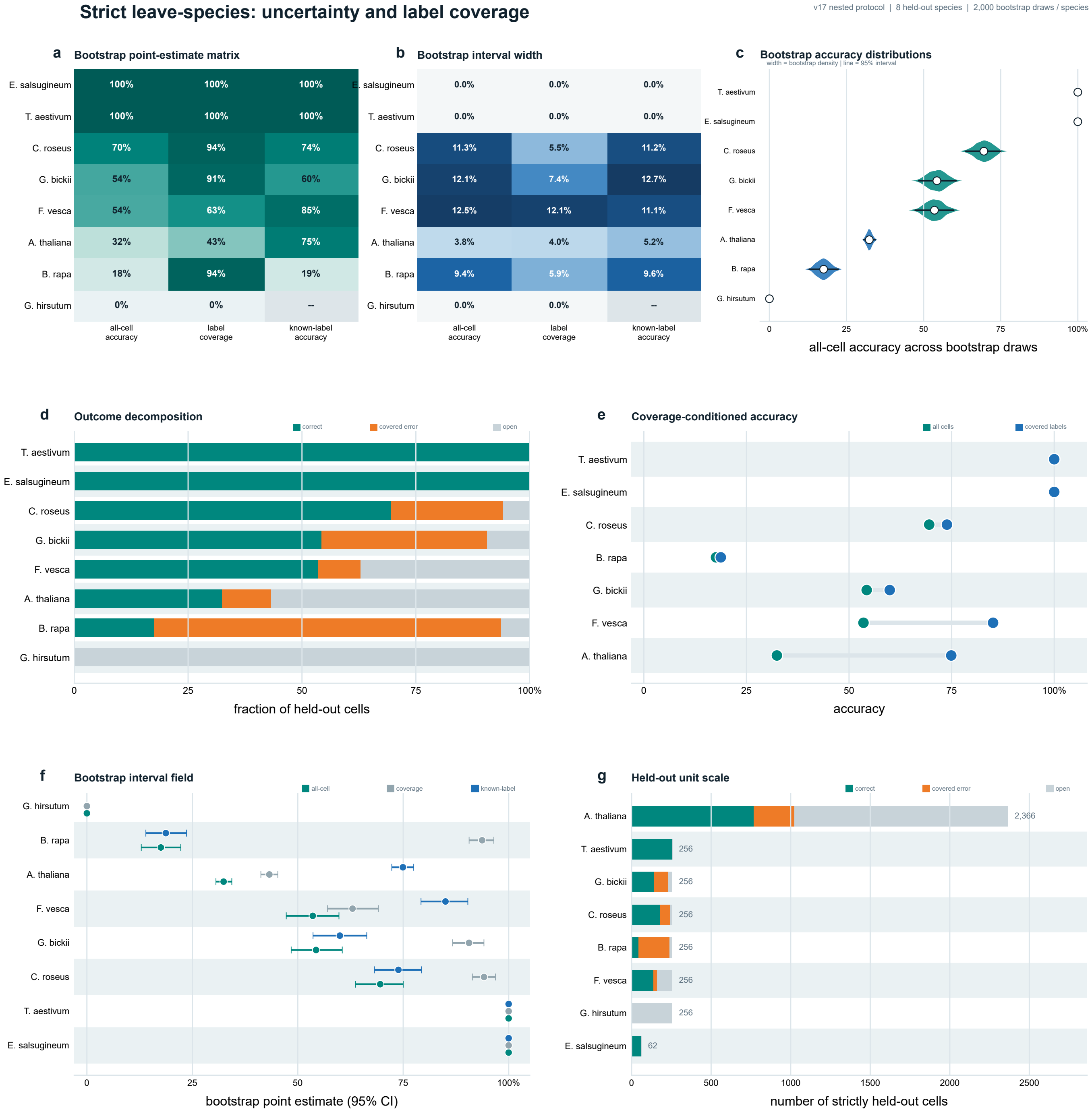



| a | all-cell accuracy | label coverage | known-label accuracy |
|---|---|---|---|
| E. salsugineum | 100% | 100% | 100% |
| T. aestivum | 100% | 100% | 100% |
| C. roseus | 70% | 94% | 74% |
| G. bickii | 54% | 91% | 60% |
| F. vesca | 54% | 63% | 85% |
| A. thaliana | 32% | 43% | 75% |
| B. rapa | 18% | 94% | 19% |
| G. hirsutum | 0% | 0% | -- |

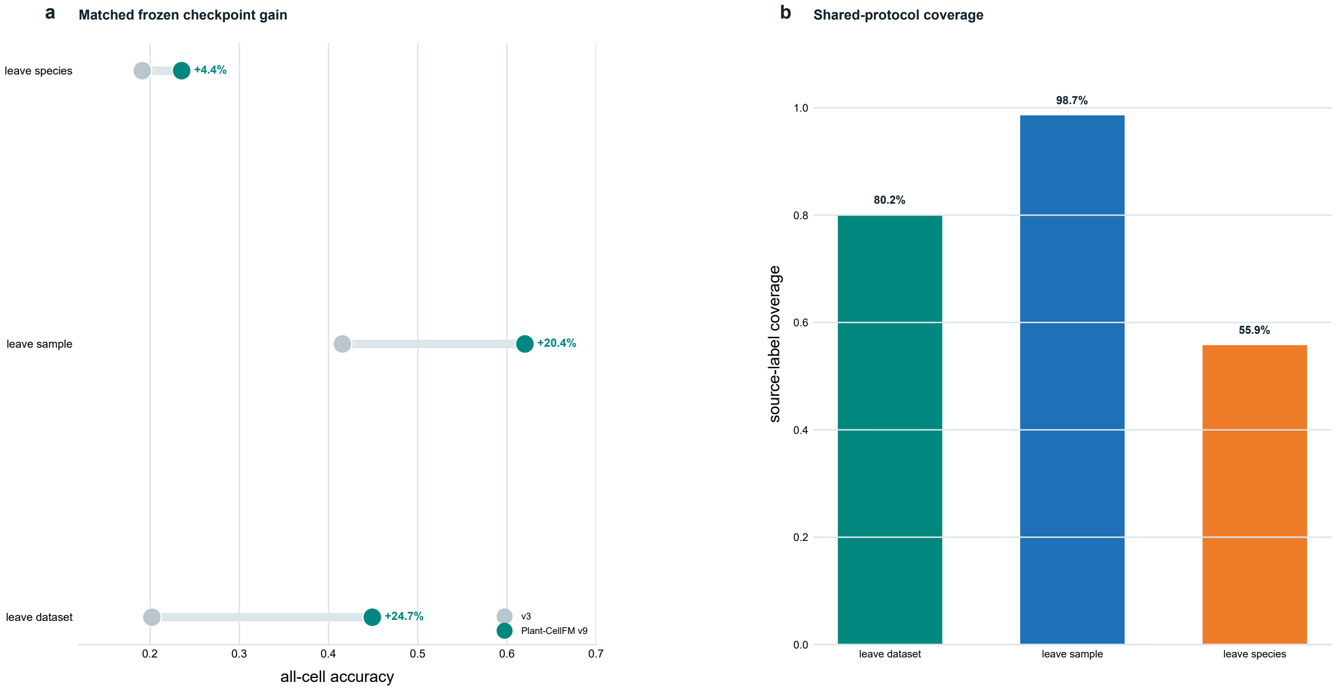
| b | all-cell accuracy | label coverage | known-label accuracy |
|---|---|---|---|
| E. salsugineum | 0.0% | 0.0% | 0.0% |
| T. aestivum | 0.0% | 0.0% | 0.0% |
| C. roseus | 11.3% | 5.5% | 11.2% |
| G. bickii | 12.1% | 7.4% | 12.7% |
| F. vesca | 12.5% | 12.1% | 11.1% |
| A. thaliana | 3.8% | 4.0% | 5.2% |
| B. rapa | 9.4% | 5.9% | 9.6% |
| G. hirsutum | 0.0% | 0.0% | -- |

| g | number of strictly held-out cells |
|---|---|
| A. thaliana | 2,366 |
| T. aestivum | 256 |
| G. bickii | 256 |
| C. roseus | 256 |
| B. rapa | 256 |
| F. vesca | 256 |
| G. hirsutum | 256 |
| E. salsugineum | 62 |


Runtime confidence and deployment boundary
separate from strict cross-species evaluation
a
Selective runtime annotation
30% accepted
96.6% accuracy
score
accepted-cell fraction
accepted-cell accuracy
rejected-error capture
b
Strict and deployment routes
strict v14
runtime head
teacher rescue t07
all-cell
known-label
macro-F1
open-set
42.4%
75.8%
30.4%
0.0%
66.2%
62.9%
34.1%
70.5%
60.1%
74.0%
34.8%
42.5%
c
Confidence threshold ledger
accepted-cell accuracy
confidence threshold
30%
40%
50%
60%
80%
100%
d
Protocol boundary
strict
held-out labels absent
runtime
full-vocabulary annotation head
rescue
confidence-gated deployment
deployment scores are not substituted for strict zero-shot scores

# Matched internal checkpoint comparison

shared data, split and label contract; not a third-party ranking



# Sorghum root: exchanged sealed-library tests

# Sorghum root: state-resolved error atlas

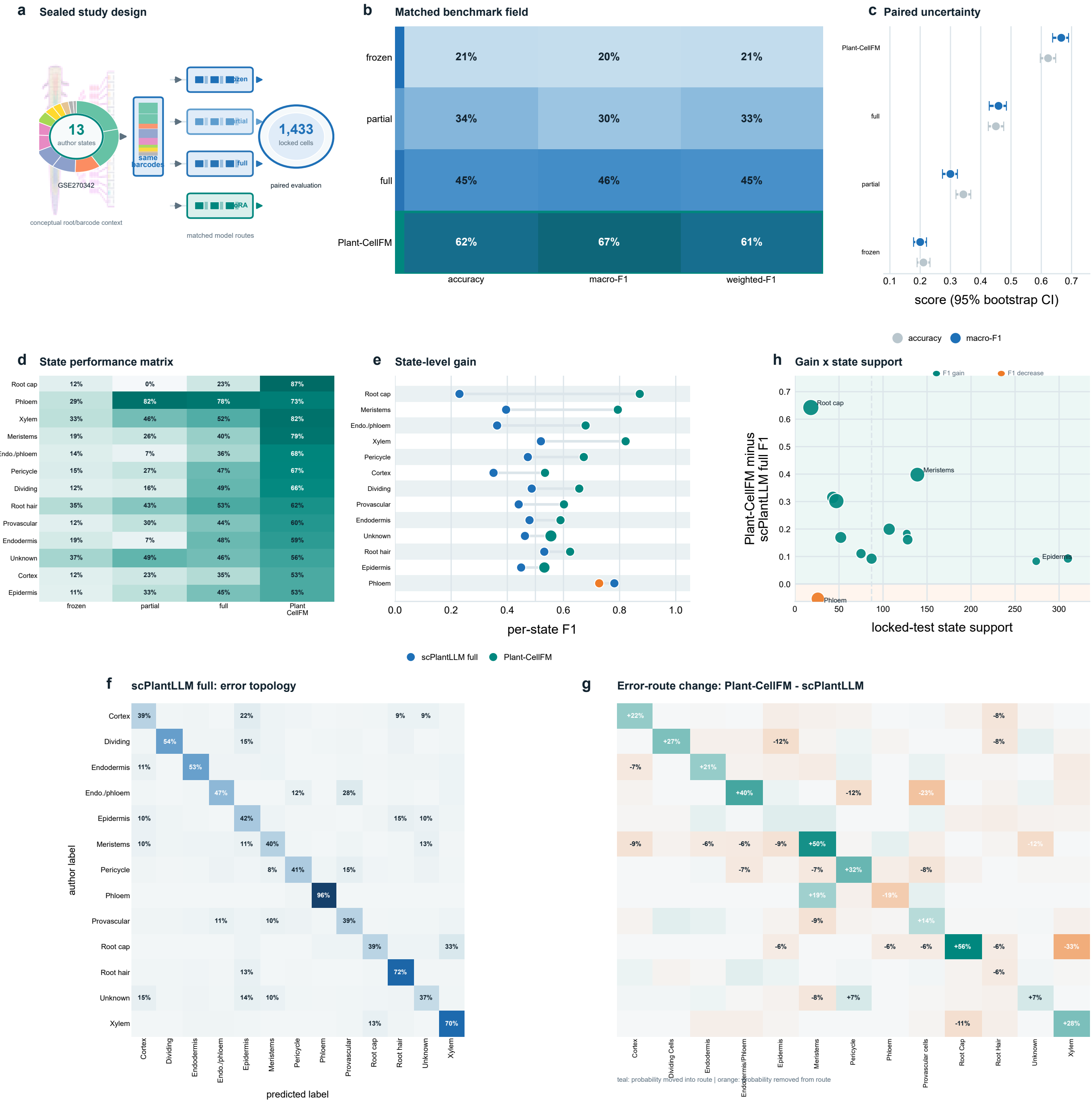
sealed OUGHW test | 4,150 cells | 27 author states

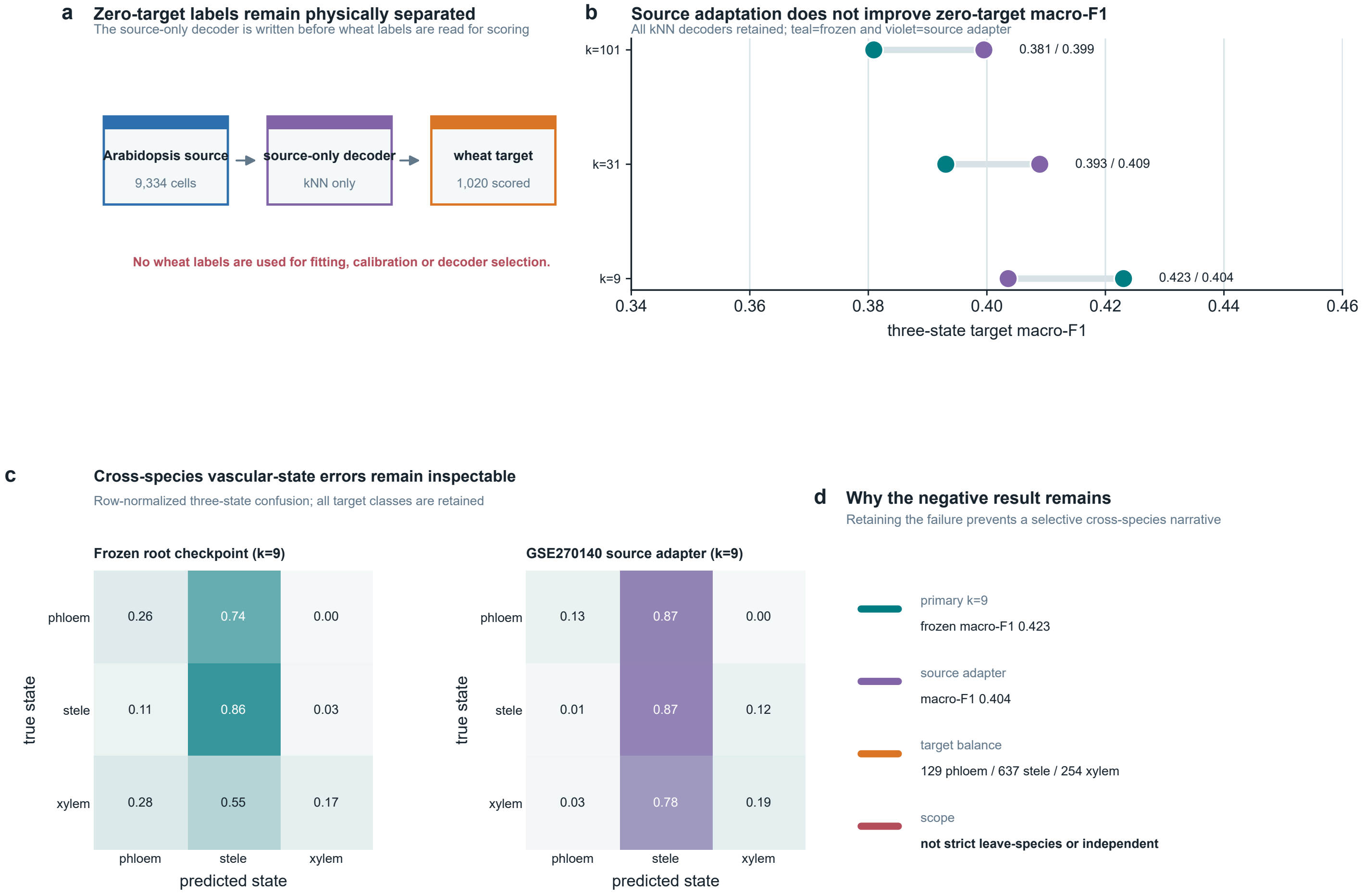
Arabidopsis root: marker-candidate programs
public data | 24 adapters | 10 root identities | 260 candidates
a Evidence route
PUBLIC MATRIX
260 candidates
PLANT-CELLFM
256-d embedding
ROOT PROGRAMS
10 identities
24 registered adapters | 200 marker rows
b Top marker programs
columella: AT5G02380, AT2G04025, AT2G36950, AT3G20840, AT3G45730
lateral cap: AT1G26820, AT3G16440, AT1G15385, AT1G06090, AT5G55110
root cap: AT1G54010, AT5G10130, AT1G28290, AT5G58784, AT2G43610
root hair: AT3G54580, AT1G30870, AT3G09925, AT3G54590, AT3G62680
non-hair: AT1G65310, AT4G12545, AT1G70850, AT1G14960, AT4G12550
cortex: AT1G12090, AT1G13930, AT1G21310, AT5G13930, AT4G30170
endodermis: AT3G22620, AT3G22600, AT2G32300, AT2G28670, AT5G15290
stele: AT4G11210, AT2G02130, AT1G12080, AT4G14130, AT3G59370
phloem: AT5G04080, AT1G62380, AT2G46630, AT1G79430, AT5G59090
xylem: AT5G03170, AT1G20850, AT5G16490, AT1G08283, AT4G23690
rank 1 rank 2 rank 3 rank 4 rank 5
candidate rank
log2 fold-change
c Marker effect geometry
detection-rate separation
marker log2 fold-change
AT3G22620
AT1G54010
AT1G65310
AT4G11210
AT5G04080
AT1G26820
AT5G03170
root identities
other states
d Identity-to-marker programs
predicted root identity
top marker candidates
columella, lateral cap, root cap, root hair, non-hair, cortex, endodermis, stele, phloem, xylem
AT5G02380, AT2G04025, AT1G26820, AT3G16440, AT1G54010, AT5G10130, AT3G54580, AT1G30870, AT1G65310, AT4G12545, AT1G12090, AT1G13930, AT3G22620, AT3G22600, AT4G11210, AT2G02130, AT5G04080, AT1G62380, AT5G03170, AT1G20850
edge width = detection separation | node radius = marker effect
e Identity-level effect summary
median log2FC | detection delta | marker score
columella 2.76 24% 0.61
lateral cap 4.21 57% 2.47
root cap 2.94 53% 1.40
root hair 3.74 40% 1.37
non-hair 2.96 52% 1.44
cortex 3.19 39% 1.13
endodermis 4.59 47% 2.21
stele 3.59 36% 1.06
phloem 5.82 35% 1.77
xylem 6.39 27% 1.77
f Rank-wise marker effect
marker log2 fold-change
candidate rank
g Identity-level effect geometry
median detection separation
median marker log2 fold-change
lateral cap, root cap, non-hair, endodermis, cortex, root hair, stele, phloem, xylem, columella


a
A context-specific state layer augments the frozen checkpoint
GSE270140/GSM8335426 author labels: 11,760 cells, 14 states, one secondary-root sample
frozen root checkpoint
13 root states
LoRA-mode adaptation
80% labelled cells
locked test report
20%: 2,352 cells
The new result is supervised adaptation, not a replacement for strict zero-shot transfer.
train: 8,232
validation: 1,176
test: 2,352
b
Labelled adaptation recovers held-out vascular states
1,885 shared-state cells; fixed-bootstrap 95% intervals
Secondary-root adapter
0.909
macro-F1 0.916
Frozen base checkpoint
0.020
macro-F1 0.060
matched three-state semantic accuracy
Base head: 2.0% | Adapter: 90.9% | +88.9 percentage points
c
All 14 secondary-root states resolve on the locked test
Row-normalized locked-test matrix; complete values in source TSV
Companion
Conductive
Fiber
Late vessel
LR primordium
Mature phloem
Mature xylem
Maturing xylem
Myrosin
Periderm
Sieve
Vascular cambium
Expanding vessel
Young xylem
predicted author label
d
Per-class F1 preserves rare-state visibility
Point area scales with test-cell support; no classes are omitted
Myrosin 0.92
Companion 0.92
Maturing xyl. 0.89
Cambium 0.88
LR prim. 0.88
Late vessel 0.88
Mature xyl. 0.87
Periderm 0.86
Mature phl. 0.85
Young xyl. 0.84
Sieve 0.81
Fiber 0.78
Expanding vsl. 0.75
Conductive 0.73
held-out per-class F1
e
Validation plateaus before locked-test evaluation
Best checkpoint is selected exclusively by validation macro-F1
validation score
selected epoch 7
validation macro-F1
validation accuracy
training epoch
f
Calibrated held-out results support review-aware use
Reported separately from strict transfer and from label-free external execution
fine accuracy 0.840
fine macro-F1 0.845
correct mean confidence 0.895
incorrect mean confidence 0.702
Test evaluator: 2,352 cells; original one-sample study design is retained in the record.

# Wheat root: matched official-checkpoint reference

GSE270342 | 1,433 locked cells | same-study adaptation, not strict transfer





a
Zero-target labels remain physically separated
The source-only decoder is written before wheat labels are read for scoring
Arabidopsis source
9,334 cells
source-only decoder
kNN only
wheat target
1,020 scored
No wheat labels are used for fitting, calibration or decoder selection.
b
Source adaptation does not improve zero-target macro-F1
All kNN decoders retained; teal=frozen and violet=source adapter
k=101
0.381 / 0.399
k=31
0.393 / 0.409
k=9
0.423 / 0.404
0.34
0.36
0.38
0.40
0.42
0.44
0.46
three-state target macro-F1
c
Cross-species vascular-state errors remain inspectable
Row-normalized three-state confusion; all target classes are retained
Frozen root checkpoint (k=9)
phloem
stele
xylem
0.26
0.74
0.00
0.11
0.86
0.03
0.28
0.55
0.17
true state
predicted state
GSE270140 source adapter (k=9)
0.13
0.87
0.00
0.01
0.87
0.12
0.03
0.78
0.19
d
Why the negative result remains
Retaining the failure prevents a selective cross-species narrative
primary k=9
frozen macro-F1 0.423
source adapter
macro-F1 0.404
target balance
129 phloem / 637 stele / 254 xylem
scope
not strict leave-species or independent

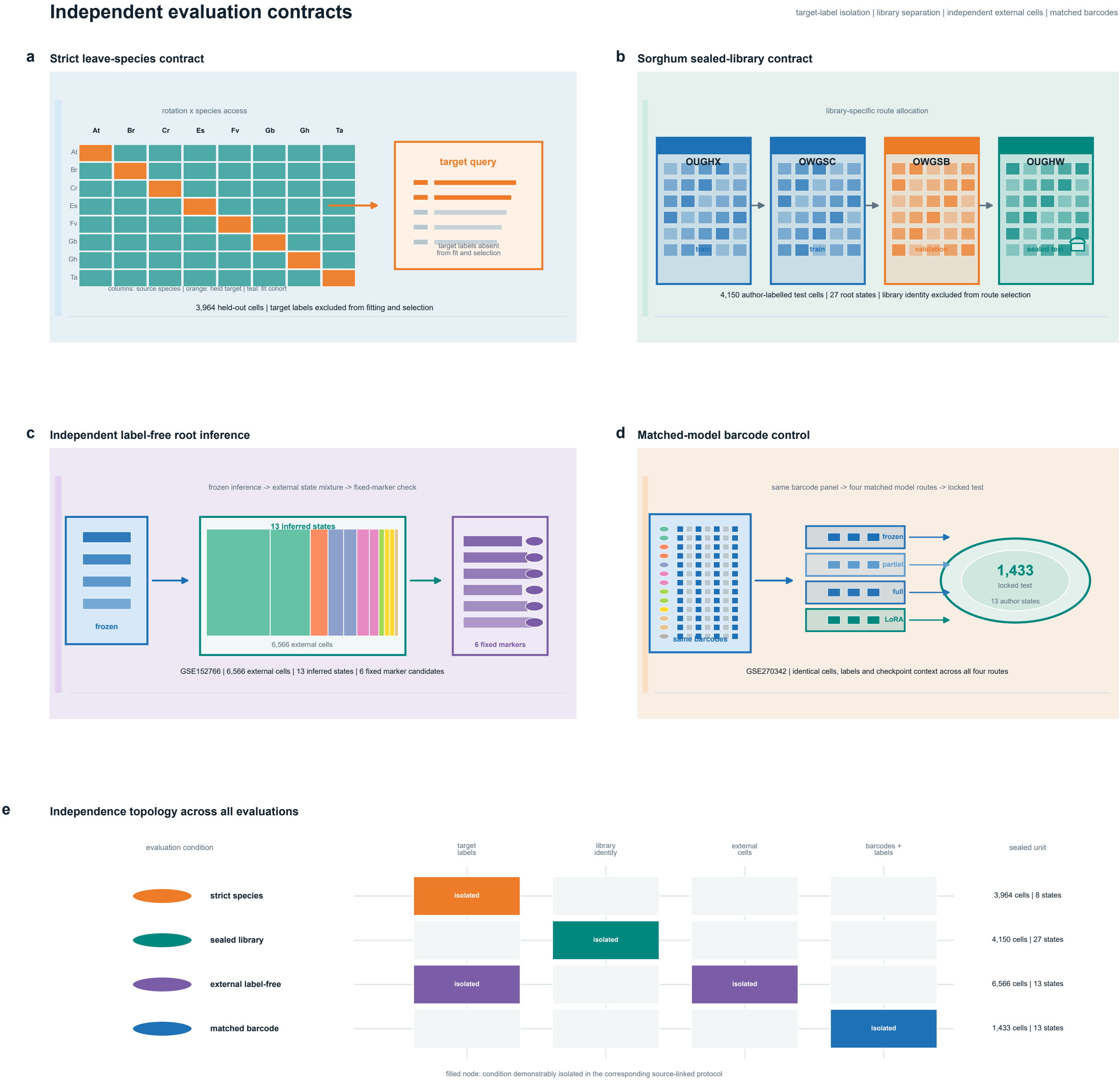
# Extended Data Figure 8 | Multi-species public-data computational case

## species summary

| species | cells | tissues | cell_types | dominant_tissue | dominant_cell_type |
|---|---|---|---|---|---|
| Arabidopsis thali | 1206 | 1 | 8 | Root tip | Non-hair |
| Gossypium hirsu | 18463 | 1 | 4 | Ovule outer integ | Outer pigment layer |
| Oryza sativa | 11443 | 1 | 13 | Pistil | Style |
| Zea mays | 391 | 1 | 5 | Pollen | Unknow |

## marker summary

| species | cell_type | n_group | top_genes | median_score | median_log2fc | median_detection |
|---|---|---|---|---|---|---|
| Arabidopsis th | G2/M phase | 233 | ['AT5G37247', | 16.266453203 | 16.154865220 | 0.0729613733 |
| Arabidopsis th | Non-hair | 254 | ['AT1G36622', | 14.154297925 | 13.824359618 | 0.0866141732 |
| Arabidopsis th | Root endoderr | 218 | ['AT4G16270', | 12.521853569 | 12.384302818 | 0.0642042013 |
| Gossypium hir | Epidermis | 5928 | ['Ghir-A12G00 | 5.8843253355 | 5.6998403976 | 0.0818095246 |
| Gossypium hir | Fiber cell | 1920 | ['Ghir-D12G01 | 4.8046585381 | 4.4687874617 | 0.2570678789 |
| Gossypium hir | Outer pigment | 9183 | ['Ghir-A04G01 | 4.0835502516 | 3.8699697135 | 0.1084327383 |
| Oryza sativa | Nucellus | 936 | ['Os01g02059 | 5.3588670173 | 4.8859142595 | 0.1664030591 |
| Oryza sativa | Outer ovary w | 2295 | ['Os03g07397 | 5.2445398383 | 5.0470232844 | 0.0999298473 |
| Oryza sativa | Style | 2367 | ['Os11g045430 | 3.9760750027 | 3.7335972201 | 0.0860737486 |
| Zea mays | G1/S phase | 97 | ['Zm00001d01 | 10.549266749 | 10.405064216 | 0.0669625477 |
| Zea mays | S phase | 77 | ['Zm00001d00 | 16.768692017 | 16.664795861 | 0.0649350649 |
| Zea mays | Unknow | 105 | ['Zm00001d01 | 11.466097363 | 11.211915196 | 0.0760312303 |



Independent evaluation contracts
target-label isolation | library separation | independent external cells | matched barcodes
a
Strict leave-species contract
rotation x species access
At
Br
Cr
Es
Fv
Gb
Gh
Ta
target query
target labels absent
from fit and selection
columns: source species | orange: held target | teal: fit cohort
3,964 held-out cells | target labels excluded from fitting and selection
b
Sorghum sealed-library contract
library-specific route allocation
OUGHX
OWGSC
OWGSB
OUGHW
train
train
validation
sealed test
4,150 author-labelled test cells | 27 root states | library identity excluded from route selection
c
Independent label-free root inference
frozen inference -> external state mixture -> fixed-marker check
frozen
13 inferred states
6,566 external cells
6 fixed markers
GSE152766 | 6,566 external cells | 13 inferred states | 6 fixed marker candidates
d
Matched-model barcode control
same barcode panel -> four matched model routes -> locked test
same barcodes
frozen
partial
full
LoRA
1,433
locked test
13 author states
GSE270342 | identical cells, labels and checkpoint context across all four routes
e
Independence topology across all evaluations
evaluation condition
target labels
library identity
external cells
barcodes + labels
sealed unit
strict species
isolated
3,964 cells | 8 states
sealed library
isolated
4,150 cells | 27 states
external label-free
isolated
isolated
6,566 cells | 13 states
matched barcode
isolated
1,433 cells | 13 states
filled node: condition demonstrably isolated in the corresponding source-linked protocol